

The key to understanding this is to ask how, exactly, one might measure the length of a moving object. We can do another thought experiment using the example of a very fast train and determine how we can measure its speed. We need a set of stationary clocks spread out at known positions. The measurement of length consists of recording the simultaneous positions of the two ends of the train – we may need multiple observers to do this. The instructions might be to report in if you see either end of the train at your location at precisely noon. We can work out, by knowing who reported in, where the ends of the train were at the chosen time.

Moving objects are length-contracted – they appear shorter to an observer in a stationary frame of reference. Note that length contraction occurs only along the direction of relative motion.

If L_0 is the length of train measured by an observer in the train and L is the length as measured by the observer in a stationary reference frame then the two measurements are related through the formula used in most textbooks:

$$\frac{L}{L_0} = \sqrt{1 - \beta^2}$$

$$L = L_0 / \gamma$$

Thoughts on length and time

Does a moving object really shrink? Reality is based on observations and measurements. If the results are always consistent and if no error can be determined, then what is observed and measured is real. In that sense, the object really does shrink. However, a more precise statement is that motion affects the measurement and thus reality. To an observer on the ground the train does shrink. In another observer's reality the amount of shrinking may be different. And for observers on the moving train the reality is that everything is as normal.

The relativity of velocities

No two objects can have a relative velocity greater than c . But what if a spacecraft traveling at $0.9c$ and it fires a missile which it observes to be moving at $0.8c$ with respect to it!? Velocities must transform according to the Lorentz transformation, and that leads to a rather non-intuitive result called Einstein velocity addition.

To analyse this situation we can consider two observers in relative motion with respect to each other who are both observing the motion of the missile. How do they measure the velocity of the object relative to each other if the speed of the object is close to that of light? We can consider a reference frame S' moving at a speed v relative to S . The missile has a velocity u' measured in the S' frame. The velocity u is the speed of the missile relative to frame S .

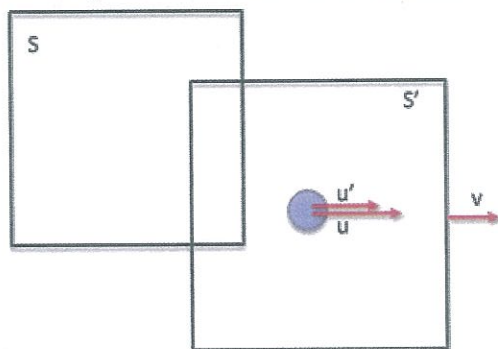


Figure 14.8: A reference frame S' is moving at a speed v relative to S . The missile shown by the blue circle has a velocity u' measured in the S' frame. The velocity u is the speed of the missile relative to frame S .

Special relativity explained

Notes

$$\text{Then } u' = \frac{u - v}{1 - \frac{uv}{c^2}}$$

$$\text{and } u = \frac{u' + v}{1 + \frac{u'v}{c^2}}$$

When v is much less than the speed of light c the denominator in the equation for u' approaches unity, and so $u' = u - v$, which is the Galilean velocity transformation equation. It is exactly what we would expect in the non-relativistic case. However, when u approaches the speed of light the equation becomes:

$$u' = \frac{c - v}{1 - \frac{cv}{c^2}} = \frac{c - v}{1 - \frac{v}{c}} = \frac{c(1 - \frac{v}{c})}{1 - \frac{v}{c}} = c$$

From this result, we see that the speed of a particle travelling close to the speed of light measured by an observer in frame S is also measured as c by an observer in S' . This result is independent of the relative motion of S and S' . This is consistent with Einstein's second postulate, that the speed of light must be c relative to all inertial reference frames. Furthermore, we find that the speed of an object can never be measured to be greater than c .

What can two observers agree on?

Two observers do agree on:

- (1) their relative speed of motion with respect to each other
- (2) the speed of any ray of light, and
- (3) the simultaneity of two events which take place at the same position and time in some frame.

We have seen several measurements that the two observers do not agree on:

- (1) the time interval between events that take place in one of the frames,
- (2) the distance between two fixed points in one of the frames,
- (3) the velocity of a moving particle, and
- (4) whether two events are simultaneous or not.

Problem Set 14.1: Special relativity explanation questions

Notes

- (a) What is meant by the term “inertial reference frames”
 - (b) Are two vehicles moving at different but constant velocities in the same inertial reference frame?
- State Einstein’s postulates in his special theory of relativity.
- The term “proper time” is used in special relativity. What is meant by this term?
- Light does not undergo any effects like length contraction and time dilation so is it subject to the theories of special relativity?
- A photon has energy but does not obey Einstein’s mass/energy equation $E = mc^2$. Explain.
- Two trains are moving at different but constant velocities. Are there any conditions under which they are in the same inertial reference frame? Do two non-inertial reference frames imply that one frame is accelerating?
- Two spaceships are launched from the same place at the same time in opposite directions, both eventually reaching a velocity of $0.70c$, in their respective directions. The technicians at the launch site see both of these spaceships as travelling away from them at $0.70c$. When the second spaceship is viewed from the first and vice versa what would the velocity of the other ship be? Numerically it would appear that the speed at which either ship is moving away from the other is $1.40c$. Discuss the validity of this supposition.
- Given a jet fighter can fly at supersonic speed (say Mach 3) does design allowance need to be incorporated to accommodate any shrinkage in length? Would this be necessary in spacecraft design if the spacecraft could travel at near light speed? Explain.
- Is the density of the material from which a relativistic space ship is constructed itself relative?
- Alpha Centauri is a star 4.367 light years from Earth. A time traveller taking this journey at near light speed in a starship was confused by the fact that it he made the journey in less than 4 years based on his clock. How would you explain this situation to the traveller? How would this dilemma be viewed by an observer on Earth?
- Explain if or how the density of a material is affected when it is travelling at relativistic velocities.
- A diagram of a beam of light bouncing off mirrors shows that time in a moving spacecraft must run slow as seen by a stationary observer.
 - (a) Explain why there must be length contraction.
 - (b) Is this contraction real or an optical illusion?
- When an object travels at near light speed we talk about the effects of time dilating relative to an observer in a different reference frame and length contracting to approach zero as its velocity approaches light speed. Why is it that light itself not subject to these two effects?
- Special relativity considers the speed of light as being constant in all reference frames. How does this explain refraction of a beam of light by a glass prism when this is said to occur because the velocity of the light has decreased?

15. You are the pilot of a spaceship and are looking for a safe runway on which to land. You see one on a planet you are passing when travelling at $0.70c$ and measure it to be 2.2 km long and 600 m wide. What are the actual dimensions of this landing strip as measured on the ground?
16. You are in a spaceship with no windows, radios or other means to check outside. How would you determine if the spaceship is at rest or moving at constant velocity?
- A. by determining the apparent velocity of light in the spaceship
 - B. by checking your precision watch- if it's running slow, then the ship is moving.
 - C. by measuring the lengths of objects in the spaceship. If they are shorter, then the ship is moving
 - D. you should give up because you've taken on an impossible task.
- Explain your choice.
17. It is said that Einstein, in his teenage years, asked the question: "What would I see in a mirror if I carried it in my hands and ran with the speed of light?" How would you answer this question?
- A. The mirror would be totally black
 - B. You would see the same thing as if you were at rest
 - C. The image would be distorted
 - D. None of the above
- Explain your choice.
18. You are riding in a spaceship travelling at $0.6c$ when you shine your headlights into the distance. Another spaceship is travelling toward you and a scientist on board measures the speed of the light beam coming from your spaceship. What will he observe?

Problem Set 14.2: Special relativity calculation questions

Notes

1. At what velocity is a spaceship travelling if a clock on it runs at a rate which is one-half the rate of a clock at rest relative to the spacecraft? 13.
2. Calculate the speed of a spaceship that is measured by an observer at rest at 0.70 of its actual length.
3. Calculate the momentum of a spacecraft with a rest mass of 2.5 tonne travelling at a velocity of 0.92c.
4. Two vehicles, both travelling at 0.80c, are moving apart from each other in exactly opposite directions. Calculate the velocity of one vehicle relative to the other.
5. A spacecraft is moving past a stationary observer at 0.92c. How long is one second of proper time on the spacecraft as measured by the stationary observer?
6. A train with a length measured at rest of 200 m is approaching a mountain tunnel of length 160 m. An observer at rest relative to the train sees that all of the train is inside the tunnel at the same time. Calculate the speed at which the train is approaching the tunnel.
7. A spacecraft is travelling to a distant galaxy 250 light years from Earth at 0.995c. How long would this spacecraft take to reach this galaxy 14
 - (a) relative to the occupants of the spacecraft?
 - (b) relative to an observer on Earth?
8. While you are on Earth, you observe that the spaceship Centauri-1 travels past you at 0.994c in 30 ns. This spacecraft is normally stored in a hanger at the local intergalactic base. What is the minimum length this hanger would need to be to completely enclose the Centauri-1? 15
9. The half-life of a π -meson (pion) is 26 μ s and it travels at a velocity of 0.95c towards Earth. What is its lifetime measured relative to a stationary point on Earth?
10. Two manned spaceships, Enterprise and Discovery, are launched so that they finish up travelling in exactly opposite directions away from each other. Enterprise has a velocity of 0.70c relative to Earth in its direction and Discovery a velocity of 0.85c relative to Earth in its direction. 16
 - (a) What velocity would an observer on Earth see each of their velocities as being?
 - (b) From the reference frame of an occupant of Discovery, at what velocity would they say the Earth was moving away from them at? 17
 - (c) An occupant of Enterprise says Discovery was moving away from them at a velocity of 1.55c. Explain why this person is wrong and give a better value for the separation velocity. You must show all equations used and calculations performed. 18
11. Our solar system is part of the Milky Way galaxy which is generally considered to have a diameter of around 1.2×10^5 light years. If a particle travelling at a velocity of 0.98c entered this galaxy, how long would it take to traverse the diameter in the reference frame of: 19
 - a) the particle
 - b) a stationary observer
12. Identical clocks are placed with you on Earth and in a spaceship travelling away from you at a velocity of 0.95c and at a particular instant they are both showing 9.00 am. When your clock shows a time of 12.00 midday, what time would the astronaut observe on his clock?

13. A space fighter, with a length of 600 m, leaves its mothership and moves away from it at a constant velocity of $0.8c$. The mothership needs to communicate with the fighter so it sends out a laser beam carrying the signal to the fighter. The crew on the fighter and the communications officer on the mothership both record zero time at the instant the laser beam arrives at the rear of the fighter.

- Can each of these frames of reference be considered inertial frames?
- How long after this would the beam reach the front of the fighter as measured by the crew on the fighter?
- How long after this would the beam reach the front of the fighter as measured by the communications officer on the mothership?

On reaching the front of the fighter the laser beam is reflected back through the fighter in the direction from which it came. The reflected laser beam is again viewed by both the crew on the fighter and the communications officer on the mothership.

- Using the initial time as when the laser beam first reached the rear of the fighter, how long would it be before it again reached the rear as measured by the crew on the fighter?
- Using the initial time as when the laser beam first reached the rear of the fighter, how long would it be before it again reached the rear as measured by the communications officer on the mothership?

14. Two identical spacecraft leave the Earth at the same velocity but in directly opposite directions. If the velocity of each spacecraft was $0.84c$, what value would an observer on Earth obtain when she calculated their relative velocities?

15. You are standing on Earth observing a dogfight between two fighter jets. One of them, flying at 3.5 times the speed of sound, fires a missile at 700 m s^{-1} when directly overhead of your position. Calculate the velocity of the missile as observed by you using each of the following.

- Classical physics principles.
- Relativity theory.

16. A radioactive nucleus travelling away from you at $0.25c$ in your frame of reference undergoes decay and emits a beta particle in the direction of its travel at $0.65c$ relative to the remaining nucleus. What is the velocity of the emitted beta particle relative to you?

17. What is the relativistic mass of a space vehicle traveling at $2.25 \times 10^8 \text{ m s}^{-1}$ if it has a rest mass of 3400 kg?

18. A train that has a proper length of 158 m and a rest mass of $8.00 \times 10^5 \text{ kg}$ is traveling at $0.925c$. What is its mass and length as measured by a stationary observer?

19. Einstein said that mass and energy are the same and as a particle's velocity increases so does its mass. At what speed would a particle be travelling for its rest mass to increase to three times its rest mass?